SYSTEMS
MANAGEMENT



Hardware Management Project Journey

Jeff Hilland *Principal Architect, Intel Corporation*

Hemal Shah *Distinguished Engineer and Architect, Broadcom Inc.*

OCP PLATINUM Member

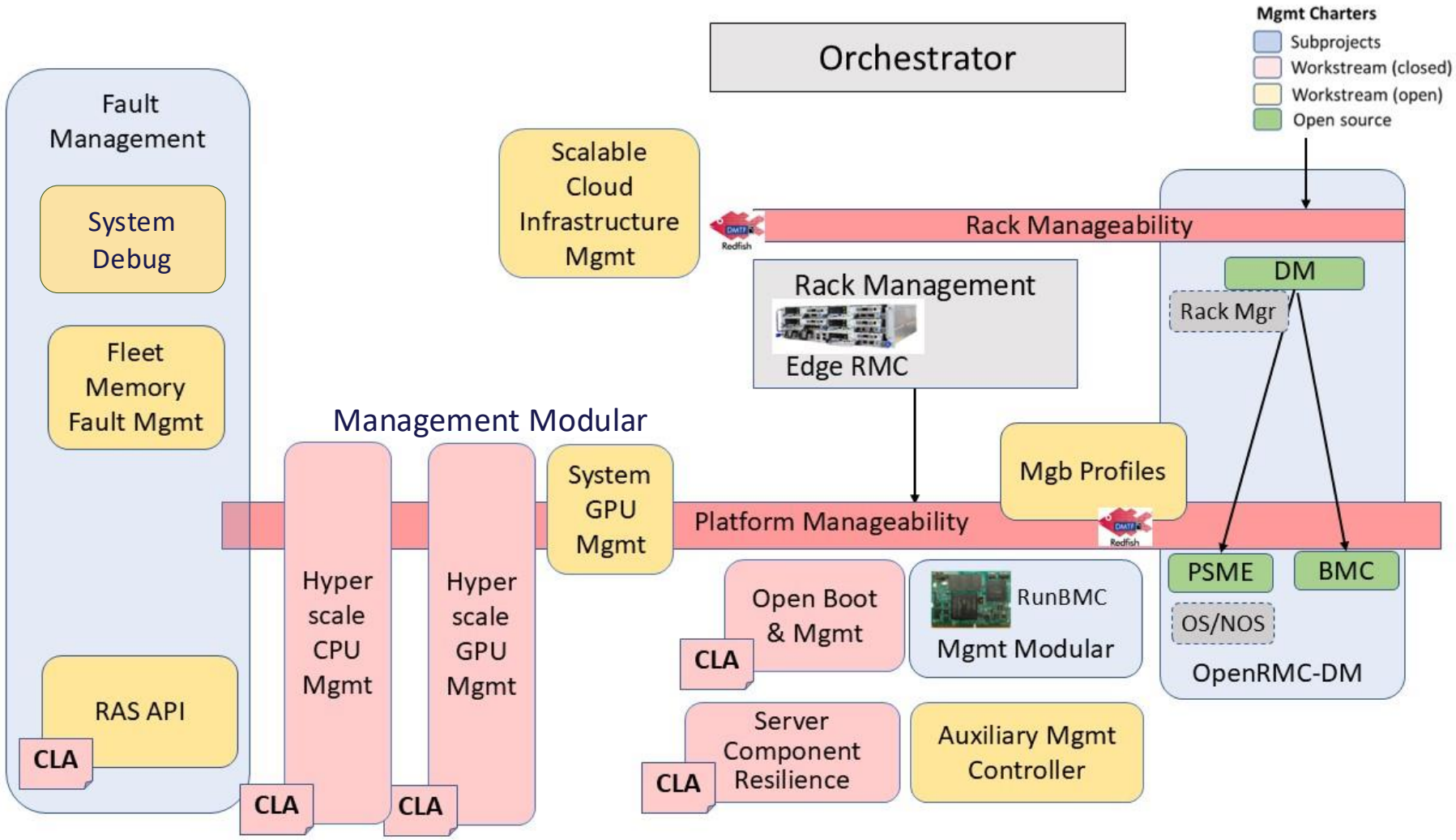
Outline

- Hardware Management Project Timeline
- Sub-Projects
  - Hardware Fault Management
  - Management Modular
  - OpenRMC-DM
- Open Workstreams
  - Auxilliary Management Controller
  - Fleet Memory Fault Management
  - Manageability Profiles
  - RAS API
  - Scalable Cloud Infrastructure Management (SCIM)
  - System Debug
  - System GPU Management
- Closed Workstreams
- Call to Action

Hardware Management Project

- Almost a decade in existence
- Started with baseline Redfish profiles for server management
- Expanded from server up to rack/data centers and down to components
- Provides manageability infrastructure spanning all OCP projects

Hardware Management Project Overview



Hardware Fault Management Sub-Project

- Defines requirements for hardware fault management
- Standardizes system behavior under hardware failures
- Defines baseline requirements for managing HW errors to achieve service levels
- Provides reference and guidance on system hardware failure management
- Includes fleet memory fault management, System Debug, and RAS API workstreams
- Current focus of the sub-project is on error injection mechanisms

Fleet Memory Fault Management

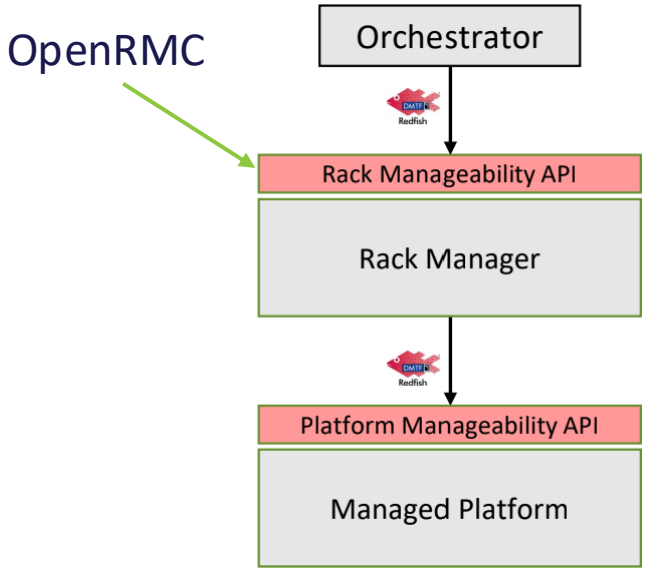
- Standardize vendor agnostic architecture for memory error handling
  - Modularization of inputs from different hardware vendors
  - APIs and connections between different modules from different vendors.
  - Define the output of each module (failure cause, health info, RAS actions..)
- Standardize memory error telemetry
  - Format content for better fleet scale RAS management
  - Troubleshooting, FRU replacement policies, etc.
- Creates Memory RAS Action codes and analyzer schema for RAS API

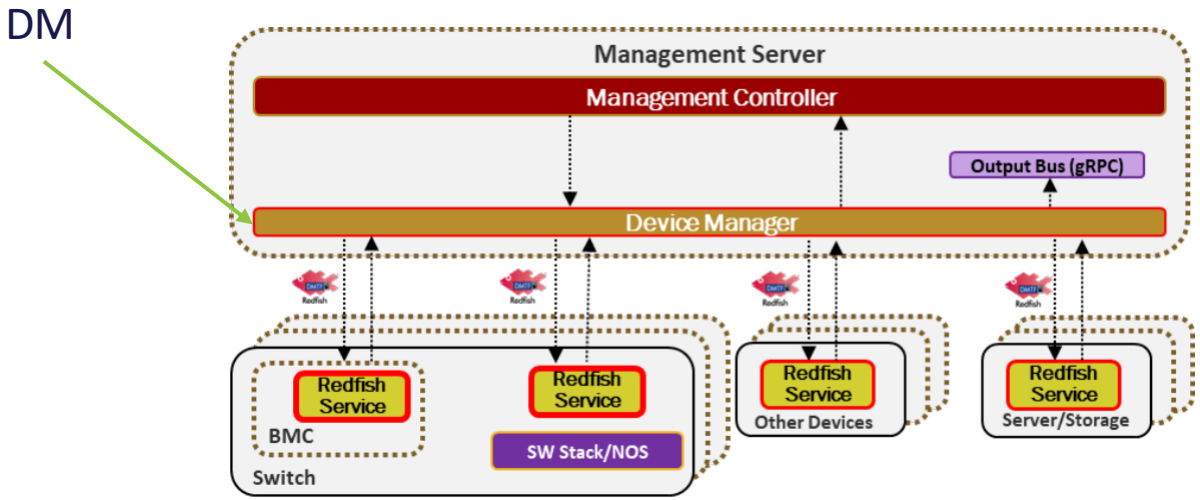
System Debug Workstream

- Previously known as the Datacenter Diagnostics and Debug Workstream
- Improve effectiveness of debugging a datacenter with different vendor components
- First spec is focused on collecting non-sensitive debug information from systems
  - Plan to be contributed by the end of the year.
- This is accomplished by creating human auditable debug data collection.
  - Code currency (what exactly is running on a given system FW versions etc.)
  - Status Dump (Information about the state of the system and its components)
- Future work includes detection of performance degradation
- To join or find out more
  - https://github.com/opencomputeproject/OCP-HM-Hardware_Management/wiki/Wiki-%E2%80%90-System-Debug

OpenRMC-DM Sub-Project

- OpenRMC specifies northbound interface for Rack Management Controller (RMC)
- OpenRMC also specifies the southbound interface requirements for the RMC
- Device Manager (DM) provides an open-source implementation of the RMC





OpenRMC-DM Update

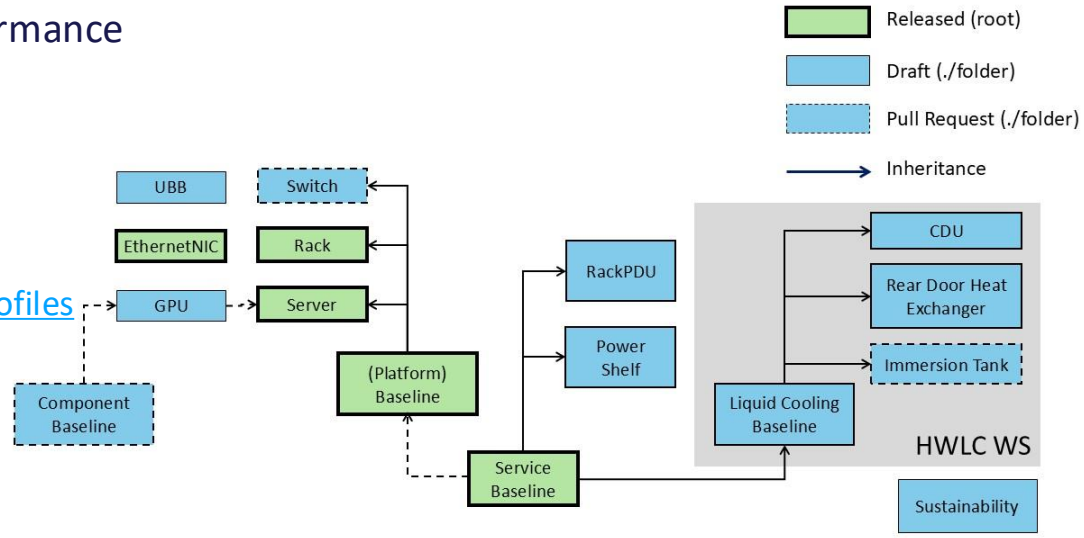
- Specification
  - Spec v1.3 kicked off focusing on Aggregation, Composability, Telemetry use cases
- Efforts on Addressing New Open Gaps
  - Physical vs logical rack scope for AI infrastructure
  - AI Driven Telemetry at Node/Rack/Pod levels
  - Increased AI Intelligence role in platform HW system management
  - Standards of additional metrics, logs, interfaces, processing engine, etc.
  - HW Management migration from reactive to proactive & predictive intelligent management
  - New challenges from disaggregated architecture over optics for next wave of AI infrastructure
- Forward-Looking Next Steps
  - New RMC/POD manager charter

System GPU Management Workstream

- Define the manageability of the GPU (off-platform and on-platform)
- Open workstream
- Consumes specifications of close Hyperscaler GPU management workstream
- Includes firmware update and serviceability
- Working on GPU Accelerator Management Interfaces Specification v1.1

Manageability Profiles

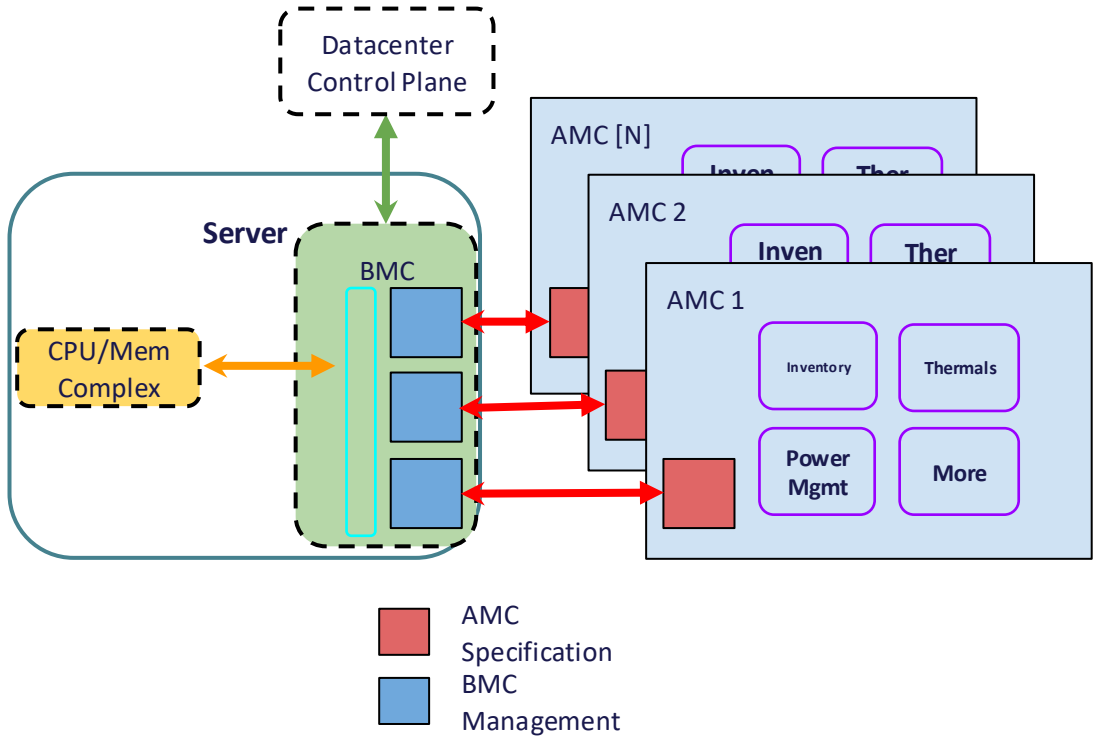
- A profile is a JSON doc which prescribes a subset of Redfish resource and properties
  - Required to be supported by an implementation for conformance
- Workstream owns
  - The baseline profile
  - Profile repository
    - https://github.com/opencomputeproject/HWMgmt-OCP-Profiles
- Platform level Profiles
  - Baseline, Server Profile, Rack Manager & Switch
- Component level Profiles
  - Component baseline, GPU, Ethernet NIC, UBB, service baseline, power shelf, RackPDU…
- Liquid Cooling Profiles
  - Liquid cooling baseline, Cooling Distribution Unit, Rear Door Heat Exchanger, Immersion Tank
- Sustainability Profile: sustainability management



https://github.com/opencomputeproject/HWMgmt-OCP-Profiles/wiki/Manageability-Profile-Workstream

Auxilliary Management Controller

- Defines the Software/Firmware API requirements for managing auxiliary devices
- AMC is a lower-cost (space, $$, power) **complement** to a BMC
  - Can be added interposer or existing HW/FW
- Target devices are aux devices to main board
  - Examples: Accelerators, Storage Controllers, Drives, NICs, DPUs & SoCs
- AMCs are not replacements for BMCs



Scalable Cloud Infrastructure Management

- Fleet management for an at-scale cloud data center (>100,000 nodes) is important
- Industry needs a comprehensive model and interfaces for at scale cloud services
- SCIM workstream
  - Specifies a standardized interface for at-scale fleet management
  - Covers phases of data center life cycle
  - Goal is to address challenges when scaling up cloud services
- Current work is focused on consolidating ideas on telemetry aggregation

Other CLA Workstreams

- Hyperscale CPU Management (closed)
- Hyperscale GPU Management (closed)
- Open Boot & Management (closed)
- RAS API (open)
  - Released two specifications
  - Actively working with DMTF on PLDM and Redfish extensions
- Server Component Resilience (closed) – no active work

Call to Action

- Hardware Management Project – Manageability interop focused
- Contribute to OCP hardware management project
- Submit proposals for specification considerations
- Submit reference architectures/products compliant to specifications
- Join hardware management project, sub-projects, and workstreams calls
- OCP Hardware Management Project
  - https://www.opencompute.org/projects/hardware-management
- OCP Hardware Management Project wiki
  - https://github.com/opencomputeproject/OCP-HM-Hardware_Management/wiki/Hardware-Mgmt-Project-Wiki
- OCP Hardware Management Specifications
  - https://www.opencompute.org/wiki/Hardware_Management/SpecsAndDesigns

Thank You!

